

12.2 The Reverse Process

Learn to denoise. Each step predicts the noise (or the clean sample) and takes a step back.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Generation walks the forward process **backward**. You start from $x_T \sim \mathcal{N}(0, \mathbf{I})$ and, step by step, remove a bit of noise until you have x_0 .

1. Predict the noise, not the image

Ho et al. (DDPM) train a net $\varepsilon_\theta(x_t, t)$ to guess the ε you added in note **12.1**. The usual loss is

$$\mathbb{E}_{t, x_0, \varepsilon} \|\varepsilon - \varepsilon_\theta(x_t, t)\|^2.$$

Predicting ε is equivalent (up to scaling) to predicting x_0 or the score $\nabla_{x_t} \log q(x_t)$. Pick one parameterization and stay consistent.

Reverse: a network predicts the noise (or x_0) and steps toward data



2. One reverse step

Given x_t and a noise prediction, a typical mean for x_{t-1} is

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \alpha_t}} \varepsilon_\theta(x_t, t) \right) + \sigma_t z,$$

with $z \sim \mathcal{N}(0, \mathbf{I})$ for $t > 1$, and σ_t from the schedule (often $\sigma_t^2 = \beta_t$).

If you **knew** x_0 , the true posterior $q(x_{t-1} | x_t, x_0)$ is Gaussian with a closed-form mean. Lab 12 uses that oracle mean for a single step so you can see denoising without training a U-Net.

3. Sampling cost

You pay T network calls (or fewer with DDIM / distilled samplers). That is the trade: stable regression training, slow ancestral sampling. GANs (Week 11) sample in one forward pass and fight mode collapse instead.

For a project, report the sampler (T , DDPM vs DDIM) next to the metric. Changing T at test time is part of the method.

4. Practice

1. If $\varepsilon_\theta = \varepsilon$ exactly, what should happen to $\|x_{t-1} - x_0\|$ relative to $\|x_t - x_0\|$, in expectation, for mid-range t ?
2. Why add $\sigma_t z$ on the way down instead of always taking the mean?
3. You may not train a score network in Lab 12. What replaces ε_θ for that one reverse step?